

MAYANK SHRIVASTAVA

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Research Interests

My research focuses on **deep generative models**. I am currently studying efficient conditional sampling from pretrained flow-matching models, with applications in **data assimilation** and probabilistic weather forecasting. More broadly, I am interested in the theoretical foundations of reinforcement learning, federated optimization, and deep learning, particularly where they intersect with generative modeling.

Relevant Coursework

Graduate-Level: Measure Theory, Probability Theory, Stochastic Calculus, Control of Stochastic Systems, Sequential Decision Making, Deep Generative Models, Statistical Reinforcement Learning

Education

University of Illinois at Urbana-Champaign

Aug. 2024 – 2028 (Expected)

Ph.D. in Computer Science

Illinois, USA

Specialization : Machine Learning

University of Illinois at Urbana-Champaign

Aug. 2022 – May 2024

Master of Science (Thesis-Track) in Computer Science

Illinois, USA

Specialization : Machine Learning, GPA: 4.0/4.0

Indian Institute of Technology Kanpur

Aug. 2016 – Aug. 2020

Bachelor of Technology, Electrical Engineering with minor in Machine Learning

Kanpur, India

GPA: 9.8/10, Department Rank : 2/135

Publications

Sketching for Distributed Deep Learning: A Sharper Analysis 

Mayank Shrivastava, Berivan Isik, Qiaobo Li, Sanmi Koyejo, Arindam Banerjee

38th Annual Conference on Neural Information Processing Systems, 2024 (NeurIPS 2024)

Beyond Johnson–Lindenstrauss: Uniform Bounds for Sketched Bilinear Forms 

Rohan Deb, Qiaobo Li, Mayank Shrivastava, Arindam Banerjee

Under review at AISTATS 2026

Grand Theft Automation: Using VLMs & Algorithmic Prompt Optimization for Theft Detection

Mayank Shrivastava, John Mckay

GenAI Workshop on Advanced Techniques, Amazon Machine Learning Conference (AMLC 2025)

Max-Quantile Grouped Infinite-Arm Bandits 

Ivan Lau, Yan Hao Ling, Mayank Shrivastava, Jonathan Scarlett

34th International Conference on Algorithmic Learning Theory, 2023 (ALT 2023)

Research Experience

Algorithmic Prompt Optimization for VLM-based Anomaly Detection

May 2025 – Aug 2025

Applied Science Summer Internship, Networks and Data Science Group

Amazon

- Developed a VLM-based change detection framework for automatic anomaly detection in warehouse photos.
- Built a two-stage pipeline using Claude VLM to describe visual changes and a text-based classifier to detect anomalies.
- Designed a prompt optimization (PO) framework where a white-box LLM generates candidate prompts.
- Applied surrogate optimization with a Neural UCB-based bandit method, improving AUC from 0.70 to 0.76.

Breaking the Dimension Dependence in Sketching for Distributed Learning

July 2023 – March 2024

Research Assistant with Dr. Arindam Banerjee

UIUC

- Analyzed the convergence rates and communication efficiency of federated learning with linear sketching.
- Existing works inherit a dependence on ambient dimensionality - the size of the deep learning model.
- For overparametrized models, we derive dimension-free convergence rates for sketching-based federated learning.
- Extended our work to differentially private settings and heterogeneous clients, showing improvements in communication.

Max-Quantile Grouped Infinite-Arm Bandits

October 2020 – March 2022

Research Assistant with Dr. Jonathan Scarlett

NUS, Singapore

- Studied the problem of max-min grouped multi-arm bandits and extended it to the case of infinite arms.
- Developed a two-phase algorithm and derived instance-dependent sample complexity bounds.

Professional Experience

AI R&D Lab, Samsung Electronics

October 2020 – March 2022

Machine Learning Engineer, Automatic Speech Recognition team

Suwon, South Korea

- Worked on training and inference engine of Conformer Speech Model for Bixby's (virtual assistant) End-to-End ASR.
- Implemented neural model for External Language Model selection (shallow fusion) - improving inference accuracy by 3%
- Implemented RIR (Impulse Response) Augmentation for improving the accuracy on far-field ASR for IoT devices.

AI R&D Lab, Samsung Electronics, South Korea

May 2019 – July 2019

Summer Internship, Automatic Speech Recognition (ASR) Team

Suwon, South Korea

- Augmented female speech dataset by implementing Voice Conversion (VC) generative models on male speech dataset.
- Implemented generative models, including Conditional VAE, and Star-GAN VC in Pytorch- improving WER (Word Error Rate) of ASR engine by 2% for female speakers through the proposed augmentation pipeline.
- Received Pre-Placement Offer from Samsung as a result of exceptional performance and sincere effort.

Indian Institute of Science, Bangalore

May 2020 – Sep. 2020

Project Assistant with Dr. Himanshu Tyagi

Bangalore, India

- Developed an agent-based simulator for modeling COVID-19 spread in Karnataka, India.
- Modelled neighborhood movement as a Markov Chain using OpenCV to extract geographical location using Maps.
- Developed sampling-based testing strategies on the simulator to reduce total tests and identify outbreak clusters.
- Developed a web application to optimally allocate collected samples across testing centers based on payload and geographical constraints.

Selected Projects

🔗 Generating Wikipedia infoboxes using LLMs | *LLMs, Vector Databases, Natural Language Processing* Dec 2023

- Developed an LLM pipeline to generate tabular summaries of Wikipedia pages over existing manual process.
- Scraped 2k+ articles to build a vector DB and proposed a novel similarity-based template retrieval system.
- Reported an improvement of 97% in BLEU scores over zero-shot prompting and comparable accuracy to ground-truth on downstream QA tasks.

🔗 Analysis of Langevin Algorithms | *Diffusion Models, Sampling, Optimization* May 2023

- Studied and implemented various algorithms for reverse diffusion in Diffusion models.
- Implemented a dimensional-independent version of Langevin Algorithm (ULA-PD) enabling dimension-free convergence.
- Proposed and evaluated a metropolis version of ULA-PD for faster convergence of neural network optimization.

🔗 Excess Food Distribution Application | *MySQL, Flask, React* Dec 2022

- Developed a web application using Flask and React linking users & restaurants for food redistribution and analytics.
- Implemented triggers, stored procedures, & indexing for superior system performance.

Technical Skills

Languages: Python, Java, C++, SQL

Technologies/Frameworks: PyTorch, MongoDB, Neo4j, MySQL, L^AT_EX, Flask, React